

Learning under Distribution Shift for Obesity Risk Prediction

Stanford CS229 Project

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1 Introduction

Distribution shifts, where the training distribution and test distribution differs, pose significant challenges to the deployment of machine learning models (Koh et al., 2021). Developing ML models that are robust to distribution shifts is crucial, particularly in high-stakes applications such as autonomous driving and medical diagnosis (Wiles et al., 2021).

In this paper, I focus on distribution shifts in supervised learning tasks. Mathematically, we aim to learn a decision rule $h : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes the loss $\mathcal{L}$ under the target distribution Q . Since Q is often unknown, the decision rule is typically learned by minimizing the expected empirical loss under the training distribution P :

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \mathbb{E}_{(X,Y) \sim P} [\mathcal{L}(h(X)), Y]$$

However, when $Q \neq P$, the estimated decision rule $\hat{h}$ might not be optimal under Q . One way to solve this problem is distributionally robust optimization (DRO), which minimizes worst-case loss over a set of plausible target distributions (also known as robustness set) (Ben-Tal et al., 2013). The optimal decision rule h^* can be formulated as

$$h^* = \arg \min_{h \in \mathcal{H}} \sup \mathbb{E}_{(X,Y) \sim Q} [L(h(X)), Y]$$

This is often an intractable optimization problem. Sahoo et al. (2022) have proposed a method called Rockafellar-Uryasev (RU) Regression within DRO framework, imposing certain constraints on the degree of distribution shift. This paper aims to compare the performance of RU Regression against a standard regression model specifically in an obesity risk prediction task, where the goal is

to predict the continuous Body Mass Index (BMI) based on demographic and health status variables. In the context of medical risk prediction, Q represents the ground truth distribution within the population, while P denotes the sampling distribution generated from Q via a selection process prone to bias. This paper alternates between bias and distribution shift, because sampling bias is the primary source of distribution shift in our analysis. Let $S \in \{0, 1\}$ denote a selection indicator. Then, it holds that $\mathbb{P}_Q(Y|X, S = 1) = \mathbb{P}_P(Y|X)$, and the selection bias is bounded as follows:

$$\frac{\mathbb{P}_Q(S = 1 | X, Y)}{\mathbb{P}_Q(S = 1 | X)} \in [\Gamma^{-1}, \Gamma]$$

for some constant $\Gamma \geq 1$. Γ characterizes the degree of selection bias, where $\Gamma = 1$ signifies no bias and larger values indicate increased flexibility to accommodate greater levels of bias.

In RU regression, hypothesis space $\mathcal{H}$ is the robustness set $\mathcal{S}_\Gamma(P, Q)$ which is set of plausible test distributions that can generate P under Γ sampling bias. It can be shown that solving optimal decision rule h_Γ^* within $\mathcal{S}_\Gamma(P, Q)$ is a convex optimization problem that can be learned using gradient descent. The details of the proof can be found in (Sahoo et al., 2022).

2 Related Work

In the work by Arletti (2024), a variant of Rockafellar-Uryasev (RU) Regression was proposed that incorporates prior knowledge regarding the direction of the bias. This approach can be beneficial assuming the correct directional bias is considered. However, such information is not always accessible.

Previous distributionally robust optimization (DRO) frameworks differ from RU regression in several ways. Esteban-Pérez and Morales (2022) constructed a robustness set by trimming the tails of the training distribution P based on a distance metric (e.g., Wasserstein distance). In contrast, RU regression defines the robustness set implicitly through the penalty parameter Γ , considering the entire training distribution P without exclusions. This implicit approach makes RU regression easier to implement and more interpretable. The work by Duchi and Namkoong (2021); Oren et al. (2019) considers distribution shifts over the joint distribution (X, Y) rather than shifts that hold conditionally on every $x \in X$. Traditional approaches for addressing distribution shift, such as covariate shift adjustment, rely on the assumption that the source of the shift is known and apply importance weighting (Sugiyama et al., 2007; Nair et al., 2019). However, these methods might perform poorly when the true target distribution is unknown or the source of the shift is complex.

3 Dataset and Features

The Behavioral Risk Factor Surveillance System BRFSS (2021-2023) serves as the training data but is prone to bias as it is collected through telephone surveys.

The National Health and Nutrition Examination Survey (NHANES) (2021-2023), which provides physically measured data, is considered the ground truth. Data is split into training (71,448 samples), validation (17,863 samples), and testing (5,169 samples). The input features are: [Gender, Age, Race, Education, Diabetes, Hypertension, Poverty_Level, Health_Insurance]. These covariates were selected based on previous study [Hopman et al., 2007; Juvanhol et al., 2016]. All input features were categorical except for Poverty_Index. Categorical data were converted using one-hot encoding. The Poverty_Index was calculated based on household income and size, as well as the corresponding federal poverty guideline for the given year. The output of the model is BMI. BMI values greater than 40 were truncated to 40, as predictions above this threshold do not add meaningful value, given that they fall within the morbid obesity range.

4 Methods

To implement RU regression, I jointly train two neural networks: one learns decision rule h that maps input features to outputs space, while the other learns an auxiliary function α that map from input features to $\mathbb{R}$. The two neural

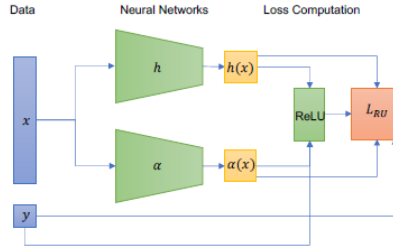


Figure 1: model architecture

networks share the same structure, consisting of two hidden layers with 64 neurons each and ReLU activation functions. The augmented loss function L_{RU}^{Γ} is defined as:

$$L_{RU}^{\Gamma} = \frac{L(h(x), y)}{\Gamma} + \left(1 - \frac{1}{\Gamma}\right) a(x) + \left(\Gamma - \frac{1}{\Gamma}\right) \text{ReLU}[L(h(x), y) - a(x)]$$

L represents the regular Mean Squared Error (MSE) loss, and Γ is a hyperparameter to be optimized.

For comparison, a neural network with same structure as well as a linear regression model, were trained. Before training, the input data points were reweighted to adjust for covariate shift. Specifically, a logistic regression model was trained on a constructed dataset, where the input features from the training set (X_{train})

were assigned a label of -1 , and the features from the testing set (X_{test}) were assigned a label of $+1$. The logistic regression model learns to distinguish between the two domains. The density ratio $w(x)$ is then estimated as:

$$w(x) = \frac{p(x \mid \text{test})}{p(x \mid \text{train})} \approx \frac{\sigma(z(x))}{1 - \sigma(z(x))},$$

where $z(x)$ is the output of the logistic regression model (logit score), and $\sigma(z)$ is the sigmoid function. The weights $w(x)$ are used to reweight the training data points.

5 Experiments / Results / Discussion

RU regression was trained using the Adam optimizer and stochastic mini-batch gradient descent for 100 epochs. A learning rate of 0.01, selected after tuning between 0.1 and 0.001, yielded the best performance. Models with varying Γ values were evaluated to identify the optimal Γ . Additionally, a regular neural network and linear model were trained with and without covariate shift adjustment. Root Mean Squared Error (RMSE) was used as the evaluation metric on the test set, as summarized in Table 1. Test and train error were plotted against different Γ , as shown in Figure 2.

Method	Train RMSE	Test RMSE
NN Model	5.27	6.61
Linear Model	5.26	6.61
NN Model(Adjusted)	5.26	6.53
Linear Model(Adjusted)	5.26	6.38
RU ($\Gamma = 1.5$)	5.60	6.17
RU ($\Gamma = 2$)	5.60	6.09
RU ($\Gamma = 3$)	5.60	6.10
RU ($\Gamma = 4$)	5.73	6.03
RU ($\Gamma = 5$)	5.63	6.05
RU ($\Gamma = 10$)	5.86	6.05

Table 1: Comparison of model performance using RMSE

RU regression models underperform during training but outperform during testing, indicating robustness to distribution shifts. The model with $\Gamma = 4$ achieved the best test performance despite poorer training results, likely due to stronger bias correction with higher Γ . However, as Γ continues to increase, performance starts to decline due to excessive bias correction and thus reduced fitting on the data.

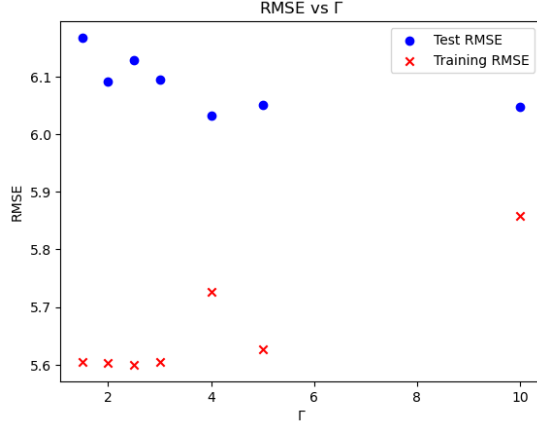


Figure 2: RMSE vs Γ

6 Conclusion / Future Work

This paper implements RU regression to address distribution shifts between training and test distributions. The experiments demonstrate that RU regression outperforms regular regression models even after adjusting for covariate shifts. This is likely due to the presence of multiple sources of distribution shifts, which is common in practice and difficult to eliminate Ward et al. (2019). Predicting individual BMI from the available data is inherently challenging, as evidenced by the large errors in all models. Subgroup analyses at the state level could offer valuable insights for localized policy-making. However, NHANES lacks state-level data. Future research could explore alternative datasets with subgroup information to enable more granular analyses. Another direction of exploration could focus on high-dimensional data, such as those from medical imaging or language tasks, to further evaluate RU regression’s capacity to handle distribution shifts in more complex scenarios.

7 Contributions

All work presented in this paper, including the implementation, experiments, and analysis, was conducted solely by the author.

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